The term Schwannoma most commonly refers to the Neurolemmoma (a benign tumor of nerve tissue origin) [1] or, less frequently, is used as a synonym for its malignant counterpart, the Malignant Peripheral Nerve Sheath Tumor (MPNST)

The origin of Schwannoma (Neurolemmoma) is accepted by most investigators to be derived from Schwann cells

1. Benign Schwannoma (Neurolemmoma)

The benign Schwannoma, also known as Neurilemmoma, perineural fibroblastoma, neurinoma, or lemmoma, is a relatively common tumor derived from Schwann cells

The neurolemmoma is typically a slowly growing lesion and usually of long duration at the time of presentation

Appearance and Location: The soft-tissue lesion is generally a *single, circumscribed nodule* of varying size that presents no pathognomonic features [8]. It may arise at any age and shows no gender predilection [7].

Central Lesions: The neurolemmoma has been reported as a central lesion within bone, primarily the *mandible, apparently originating from the mandibular nerve [8]. These central lesions may cause considerable **destruction of bone with expansion of the cortical plates*, potentially resembling a more serious lesion [8].

Symptoms: Pain and paresthesia may accompany central lesions of bone [8].

Association: The lesion does occur with some frequency in patients with neurofibromatosis [7].

Histologic Features

The microscopic appearance of the neurolemmoma is characteristic and usually cannot be confused with other lesions

1. The tumour is classically described as being composed of two types of tissue:
- **Antoni type A tissue**: Characterized by cells with elongated or spindle-shaped nuclei aligned to form a characteristic palisading pattern. The intercellular fibres are arranged in parallel fashion between rows of nuclei, sometimes appearing in whorls or swirls
- **Antoni type B tissue**: Exhibits a disorderly arrangement of cells and fibers with areas resembling edema fluid and forming microcysts
2. Additionally, **Verocay bodies**, which are small hyaline structures, are characteristically present in this tumor. In nearly all instances
3. The tumor is encapsulated.
4. Importantly, unlike the neurofibroma, neurites are generally not a component of the neurolemmoma, though they may be found on its surface

2. Malignant Schwannoma (Malignant Peripheral Nerve Sheath Tumor - MPNST)

Malignant Peripheral Nerve Sheath Tumor (MPNST) is the preferred name for this spindle cell malignancy of peripheral nerve Schwann cells. It is also known as malignant neurilemmoma, neurogenic sarcoma, or neurofibrosarcoma

Histologic Features

MPNST resembles fibrosarcoma in its overall organization

Cellular Morphology: The spindled lesional cells demonstrate the characteristic wavy or comma-shaped outline and nuclear contour of Schwann cells. The cells show marked pleomorphism (variation in size and shape) and usually high mitotic activity.

Pattern: Spindle cells are arranged in sweeping fascicles interspersed with hypocellular and myxoid regions

Variants: MPNST can be classified into categories including epithelioid, mesenchymal, or glandular characteristics

Triton tumor: A specialized diagnostic term used when MPNST lesions show rhabdomyoblastic differentiation

Glandular MPNST: Contains areas with well-differentiated ductal structures

Differential Diagnosis: Differentiating MPNST from a benign nerve sheath tumor relies primarily on the presence or absence of mitotic activity. It must also be distinguished from leiomyosarcoma (which has more distinct eosinophilic cytoplasm and a blunted nucleus) and monophasic synovial sarcoma (rare in the oral cavity)